

Aerospace series**Clad sheets and strips****Aluminium alloy 2024 Standard surface quality****General (No Code) and Specific (S) tolerances****Thickness $0,2\text{mm} \leq a \leq 6,0\text{mm}$** **Dimensions**

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1 Scope

This standard specifies the dimensions general and specific tolerances of clad sheets and strips with standard surface quality in aluminium alloy 2024, thickness $0,2\text{mm} \leq a \leq 6,0\text{mm}$, for use in aerospace applications.

The products made out of F-Temper (code B) and O-Temper (code C) are in chemical milling quality; T351-Temper products (code A) are in chemical milling quality. T3-Temper products (code D) are not in chemical milling quality.

2 Normative references

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

- AIMS03-04-000
- Technical specification - Aluminium and aluminium alloy wrought products - sheet, plate and strip up to 12 mm.
- AIMS03-04-009
- Aluminium alloy (2024), Solution treated, controlled stretched and naturally aged (T351), Clad sheet and strip $0,8 \leq a \leq 6\text{mm}$, Standard surface quality
- AIMS03-04-010
- Aluminium alloy (2024), Solution treated and naturally aged (T42), Clad sheet and strip $0,2 \leq a \leq 6\text{mm}$, Standard surface quality
- AIMS03-04-014
- Aluminium alloy (2024), Solution treated and naturally aged (T3), Clad sheet and strip $0,2 \leq a \leq 6\text{mm}$, Standard surface quality
- EN3848
- Aerospace series - Aerospace series; semi finished metallic products; Methods of measuring for form deviations.¹

3 Form

See figure 1.

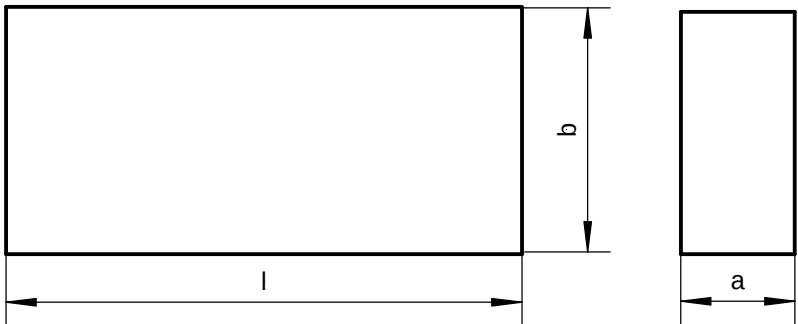


Figure 1:

¹ Published as ASD Prestandard at the date of publication of this standard

4 Recommended dimensions and mass

See table 1.

Table 1:

Dimensions in mm

Size code	a ^{1) 4)} Nominal mm	Normal format ¹⁾ b x l mm x mm	Mass per unit area ²⁾ kg/m ²
002	0,2 ³⁾	1250 x 2500	0,56
003	0,3 ³⁾		0,84
004	0,4 ³⁾		1,12
005	0,5 ³⁾		1,40
006	0,6 ³⁾		1,68
008	0,8 ³⁾		2,24
010	1,0		2,80
012	1,2		3,35
014	1,4		3,92
015	1,5		4,20
016	1,6		4,48
018	1,8		5,04
020	2,0		5,60
021	2,1		5,88
022	2,2		6,16
023	2,3		6,44
024	2,4		6,72
025	2,5		7,00
028	2,8		7,84
030	3,0		8,40
032	3,2		8,95
036	3,6		10,08
040	4,0		11,20
042	4,2		11,76
044	4,4		12,32
050	5,0		14,00
060	6,0		16,80
1) Other thickness, width and length on request 2) For information, calculated with a density: 2,8 kg/dm³ 3) Thickness is not suitable for chemical milling. 4) Strips are available up to 3,2 mm.			

5 Tolerances

5.1 Dimensional tolerances

5.1.1 Thickness

The thickness tolerances may be:

- General: No Code
- Specific: Code letter S

See table 2.

Table 2:

Dimensions in mm

Dimensions in mm

Thickness	Tolerances for width:											
	b ≤1000 ¹⁾		1000 < b ≤1250		1250 < b ≤1600		1600 < β ≤ 2000		2000 < b ≤2438		2438 < b ≤3350	
	No Code	Code S	No Code	Code S	No Code	Code S	No Code	Code S	No Code	Code S	No Code	Code S
0,2 ≤ a ≤ 0,4	± 0,04	+0,08 0	± 0,05	+0,10 0	± 0,06	+0,12 0	–	–	–	–	See relevant IPS	See relevant IPS
0,4 < a ≤ 0,6	± 0,05	+0,10 0	± 0,06	+0,12 0	± 0,07	+0,14 0	± 0,08	+0,16 0	–	–		
0,6 < a ≤ 0,8	± 0,06	+0,12 0	± 0,07	+0,14 0	± 0,08	+0,16 0	± 0,10	+0,20 0	–	–		
0,8 < a ≤ 1,0	± 0,06	+0,12 0	± 0,07	+0,14 0	± 0,09	+0,18 0	± 0,12	+0,24 0	–	–		
1,0 < a ≤ 1,2	± 0,07	+0,14 0	± 0,08	+0,16 0	± 0,10	+0,20 0	± 0,14	+0,28 0	–	–		
1,2 < a ≤ 1,4	± 0,07	+0,14 0	± 0,08	+0,16 0	± 0,10	+0,20 0	± 0,16	+0,32 0	–	–		
1,4 < a ≤ 1,6	± 0,07	+0,14 0	± 0,08	+0,16 0	± 0,12	+0,24 0	± 0,16	+0,32 0	±0,16	+0,32 0		
1,6 < a ≤ 1,8	± 0,08	+0,16 0	± 0,10	+0,20 0	± 0,12	+0,24 0	± 0,16	+0,32 0	± 0,18	+0,36 0		
1,8 < a ≤ 2,2	± 0,08	+0,16 0	± 0,10	+0,20 0	± 0,12	+0,24 0	± 0,18	+0,36 0	± 0,22	+0,44 0		
2,2 < a ≤ 2,8	± 0,10	+0,20 0	± 0,12	+0,24 0	± 0,14	+0,28 0	± 0,18	+0,36 0	± 0,25	+0,50 0		
2,8 < a ≤ 3,4	± 0,12	+0,24 0	± 0,14	+0,28 0	± 0,15	+0,30 0	± 0,20	+0,40 0	± 0,27	+0,54 0		
3,4 < a ≤ 4,0	± 0,14	+0,28 0	± 0,16	+0,32 0	± 0,20	+0,36 0	± 0,25	+0,50 0	± 0,27	+0,60 0		
4,0 < a ≤ 5,0	± 0,16	+0,32 0	± 0,20	+0,40 0	± 0,25	+0,50 0	± 0,25	+0,60 0	± 0,30	+0,70 0		
5,0 < a ≤ 6,0	± 0,20	+0,40 0	± 0,25	+0,50 0	± 0,25	+0,60 0	± 0,25	+0,70 0	± 0,30	+0,80 0		
1) For strips up to 2 mm thickness the actual thickness within one coil (length / width) should not be exceed by more than 0,04 mm. The respective limit deviations should not be exceed thereby.												

5.1.2 Width

See table 3.

Table 3:

Dimensions in millimetres

Width	Tolerances
All	+10 0

5.1.3 Length

See table 4.

Table 4:

Dimensions in millimetres

Length	Tolerances for all thickness:
$l \leq 5000$	+10 0
$l > 5000$	$+0,002 \times l$ 0

5.2 Geometric tolerances (not valid for coiled strips)

5.2.1 Squareness

5.2.1.1 Method of measurement and symbols

See EN3848

5.2.1.2 Tolerances

See table 5.

Table 5:

Dimensions in millimetres

Length	Maximum difference in the length of diagonals for all width and for thickness $0,2 < a \leq 6$
$l \leq 2000$	+ 8 0
$2000 < l \leq 5000$	+ 10 0
$l > 5000$	$+ 0,002 \times l$ 0

5.2.2 Lateral curvature

5.2.2.1 Method of measurement and symbols

See EN3848

5.2.2.2 Tolerances

See table 6.

The lateral curvature may be concave or convex.

Table 6:

Dimensions in millimetres

Thickness	Lateral curvature F on :	
	width	length
$0,2 < a \leq 6$	$\leq 0,002 \times b$	$\leq 0,002 \times l$

5.2.3 Flatness

5.2.3.1 Method of measurement and symbols

See EN3848

5.2.3.2 Tolerances

See table 7.

Table 7:

Dimensions in millimetres

Thickness	Flatness deviation f ¹⁾ for :		
	width	length	chord w ≥ 300
$0,2 \leq a \leq 3$	$0,005 \times b$	$0,004 \times l$	$0,005 \times w$ ²⁾
$3 < a \leq 6$	$0,004 \times b$	$0,003 \times l$	$0,0035 \times w$ ²⁾
1) The values do not apply to materials in conditions O and F for which the wave length shall not be less than 300mm.			
2) The maximum acceptable deviation for total length and width shall not be exceeded.			

6 Material

See table 8

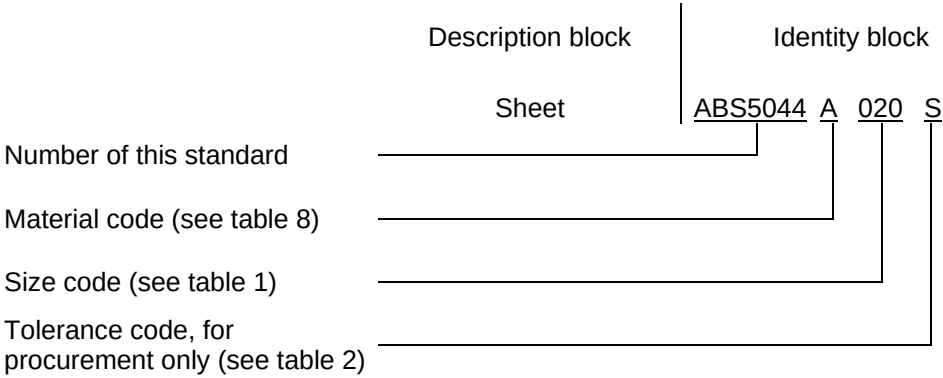
Table 8:

Dimensions in millimetres

Material						Chemical ¹⁾ milling
delivered				use		
AIMS spec. no.	condition of treatment	code	AIMS spec. no.	condition of treatment	code	
03-04-009	T351	A	03-04-009	T351	A	Yes
03-04-010	F	B	03-04-010	T42	E	Yes ²⁾
	O	C				no
	F	H				
	O	G				
	03-04-014	T3				D
1) Chemical milling suitable for thickness a > 0,8; 2) 2) No control of residual stress (as in 03-04-009 Line57)						

7 Designation

EXAMPLE:



8 Technical specification

In accordance with AIMS03-04-000

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
4	Table 1	Thickness 2,1mm; 4,2mm and 4,4 mm added
5		– General and specific tolerances added – Width: 2500/300mm added – Table 2 “Thickness tolerances” has been modified
6 12/02	Table 1 Table 8	– Thickness 3,6mm has been added (Airbus-F request) – For AIMS03-04-010 the chemical version (Code letter H and G) added (SONACA request) – Chemical milling requirements modified (Airbus-F request)
7 02/03	Clause1 and Table 8 Record of revision Issue6 Draft1	- "Suitable for chemical milling replaced by "Yes²⁾" including footnote. – For AIMS03-04-010 the "no" chemical version (Code letter H and G) added (SONACA request)
8 07/09		Tolerances for the couple with / thickness $2000 < b \leq 2438$ / $1,4 < a \leq 1,6$ added. The tolerances are the same as specified in ABS5043